MA4261
Differential Geometry of Curves and Surfaces

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3.1 Curvature of Surfaces

In Chapter 1, we defined curvature as $\kappa = |\alpha''(s)|$ — curvature **at a point.** In this chapter, we try to find a definition for curvature of a **surface** *at a point.*

Gauss had this idea: given a point $p \in S$ on a regular surface S, along the normal vector $\mathbf{N}$ perpendicular to the surface at $\mathbf{p}$, we take a knife and cut the surface downward along the tangent direction $\mathbf{v}$. The curvature of the resulting cross-sectional curve gives us information about the curvature of the surface at $\mathbf{p}$.

3.2 Linear Algebra Review

First, we define a *self-adjoint* linear transformation.

Definition 3.1 (Self-adjoint map)**.** Let V be an inner product space, and $T : V \to V$ a linear transformation. Then T is **self-adjoint** if

$$\langle T(\mathbf{v}), \mathbf{w}\rangle = \langle \mathbf{v}, T(\mathbf{w})\rangle \quad \text{for all } \mathbf{v}, \mathbf{w} \in V.$$

Lemma 3.2 (Self-adjoint via basis)**.** *Suppose we have a basis $\mathcal{B} = \{\mathbf{x}_1, \mathbf{x}_2\}$ of V. Then it suffices to check*

$$\langle T(\mathbf{x}_1), \mathbf{x}_2\rangle = \langle \mathbf{x}_1, T(\mathbf{x}_2)\rangle.$$

The usefulness of this lemma is that if we *do* have a basis, we only need to check self-adjointness on the basis vectors rather than all $\mathbf{v}, \mathbf{w} \in V$.

Definition 3.3 (Matrix representation of a linear transformation)**.** Let $\mathcal{B} = \{\mathbf{x}_1, \mathbf{x}_2\}$ be a basis of V, and $T : V \to V$ a linear transformation. Suppose

$$T(\mathbf{x}_1) = a\mathbf{x}_1 + b\mathbf{x}_2, \qquad T(\mathbf{x}_2) = c\mathbf{x}_1 + d\mathbf{x}_2.$$

Then the matrix representation of T with respect to $\mathcal{B}$ is

$$[T]_{\mathcal{B},\mathcal{B}} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Lemma 3.4 (Self-adjointness via matrix representation)**.** *Let $\mathcal{B} = \{\mathbf{x}_1, \mathbf{x}_2\}$ be an* orthonormal *basis of V. Then $T : V \to V$ is self-adjoint if and only if*

$$[T]_{\mathcal{B},\mathcal{B}} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

is symmetric, i.e. $b = c$.

Proof. Let $[T]_{\mathcal{B},\mathcal{B}} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Then

$$\langle T(\mathbf{x}_1), \mathbf{x}_2\rangle = \langle a\mathbf{x}_1 + b\mathbf{x}_2, \mathbf{x}_2\rangle = b, \qquad \langle \mathbf{x}_1, T(\mathbf{x}_2)\rangle = \langle \mathbf{x}_1, c\mathbf{x}_1 + d\mathbf{x}_2\rangle = c.$$

Hence $\langle T(\mathbf{x}_1), \mathbf{x}_2\rangle = \langle \mathbf{x}_1, T(\mathbf{x}_2)\rangle \iff b = c$, which proves both directions. $\square$

Lemma 3.5 (Existence of eigenvalue pairs)**.** *Let $A = \begin{pmatrix} a & b \\ b & d \end{pmatrix}$ be a real 2×2 symmetric matrix. The characteristic polynomial of A is*

$$p(\lambda) = \det(\lambda I - A) = \lambda^2 - (a + d)\lambda + (ad - b^2),$$

with real eigenvalues

$$\lambda_1, \lambda_2 = \frac{a + d}{2} \pm \sqrt{\left(\frac{a - d}{2}\right)^2 + b^2} \in \mathbb{R}.$$

The corresponding eigenvectors are found via

$$E_{\lambda_i} = \mathrm{Null}(\lambda_i I_2 - A) = \{\mathbf{v} \in \mathbb{R}^2 \mid (\lambda_i I_2 - A)\mathbf{v} = \mathbf{0}\}.$$

Let $\mathbf{v}_1$ be the unit eigenvector for λ_1, and let $\mathbf{v}_2$ be the unit vector perpendicular to $\mathbf{v}_1$. Then $\mathbf{v}_2$ is an eigenvector of A with eigenvalue λ_2.

Proof. Since $A = A^\top$,

$$(A\mathbf{v}_2) \cdot \mathbf{v}_1 = \mathbf{v}_2^\top A^\top \mathbf{v}_1 = \mathbf{v}_2^\top (A\mathbf{v}_1) = \lambda_1 \mathbf{v}_2^\top \mathbf{v}_1 = 0.$$

Hence $A\mathbf{v}_2 \perp \mathbf{v}_1$, so $A\mathbf{v}_2$ is parallel to $\mathbf{v}_2$, i.e. $A\mathbf{v}_2 = \lambda_2 \mathbf{v}_2$. □

Corollary 3.6 (Self-adjoint maps admit an orthonormal eigenbasis)**.** *Given a real symmetric* 2×2 *matrix, there exists an orthonormal basis of real eigenvectors*

$$\left\{ \begin{pmatrix} p_{11} \\ p_{21} \end{pmatrix}, \begin{pmatrix} p_{12} \\ p_{22} \end{pmatrix} \right\}$$

such that $P = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix}$ *is an orthogonal matrix, i.e.* $P^\top = P^{-1}$.

Theorem 3.7 (Main theorem on self-adjoint matrices)**.** *Given a real symmetric matrix* A*, we can diagonalise*

$$A = PDP^{-1} = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix} \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix}^{-1}.$$

3.3 The Gauss Map

Consider a regular surface S and a point $\mathbf{q} \in S$. We can choose a unit vector $N(\mathbf{q})$ perpendicular to the tangent plane $T_{\mathbf{q}}(S)$:

$$N(\mathbf{q}) = \frac{\mathbf{x}_u \wedge \mathbf{x}_v}{|\mathbf{x}_u \wedge \mathbf{x}_v|}, \qquad \text{or} \qquad N(\mathbf{q}) = -\frac{\mathbf{x}_u \wedge \mathbf{x}_v}{|\mathbf{x}_u \wedge \mathbf{x}_v|}.$$

Since $N(\mathbf{q}) \in S^2$ for each $\mathbf{q}$, the Gauss map sends $\mathbf{q} \mapsto N(\mathbf{q})$.

Definition 3.8 (Gauss Map)**.** Let S be a regular surface. The **Gauss map** is a smooth map

$$N : S \to S^2$$

where $N(\mathbf{q})$ is the unit normal at $\mathbf{q}$ as above.
Remarks.

1. Not every regular surface admits a Gauss map.
2. $N(\mathbf{q})$ can always *locally* be found via the cross product of coordinate vectors, which exists by regularity of S.
3. If $N : S \to S^2$ is a Gauss map, then $-N : S \to S^2$ is also a Gauss map, corresponding to the other consistent choice of unit normal.

3.4 The Second Fundamental Form

Consider the map $dN : T_p(S) \to T_{N(p)}(S^2)$. Since $\mathbf{q}$ is the origin of $T_{\mathbf{q}}(S)$ and $N(\mathbf{q})$ is the origin of $T_{N(\mathbf{q})}(S^2)$, both tangent planes contain the origin and are perpendicular to $N(\mathbf{q})$. Hence the two tangent spaces coincide, and we may view dN as a linear transformation

$$dN : T_p(S) \to T_p(S).$$

Proposition 3.9. *dN is a self-adjoint linear transformation.*

Proof. By the basis lemma, it suffices to show $\langle dN(\mathbf{x}_u), \mathbf{x}_v \rangle = \langle \mathbf{x}_u, dN(\mathbf{x}_v) \rangle$.
Since $N \circ \mathbf{x} \perp \mathbf{x}_u$, we have $\langle N \circ \mathbf{x}, \mathbf{x}_u \rangle = 0$. Differentiating with respect to v:

$$\left\langle \frac{\partial}{\partial v}(N \circ \mathbf{x}),\ \mathbf{x}_u \right\rangle + \langle N \circ \mathbf{x},\ \mathbf{x}_{uv} \rangle = 0.$$

Since $\frac{\partial}{\partial v}(N \circ \mathbf{x}) = dN(\mathbf{x}_v)$, this gives $\langle dN(\mathbf{x}_v), \mathbf{x}_u \rangle = -\langle N \circ \mathbf{x}, \mathbf{x}_{uv} \rangle$. Interchanging u and v: $\langle dN(\mathbf{x}_u), \mathbf{x}_v \rangle = -\langle N \circ \mathbf{x}, \mathbf{x}_{vu} \rangle$.
Since $\mathbf{x}_{uv} = \mathbf{x}_{vu}$, we conclude

$$\langle dN(\mathbf{x}_u), \mathbf{x}_v \rangle = -\langle N \circ \mathbf{x}, \mathbf{x}_{uv} \rangle = \langle \mathbf{x}_u, dN(\mathbf{x}_v) \rangle.$$ □

Definition 3.10 (Second Fundamental Form)**.** For $\mathbf{v} \in T_p(S)$, the **second fundamental form** of S at $\mathbf{q}$ is

$$\mathrm{II}_{\mathbf{q}}(\mathbf{v}) = -\langle dN(\mathbf{v}), \mathbf{v}\rangle.$$

Proposition 3.11 (Finding dN from II_p)**.** *Let $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$ be an orthonormal basis of $T_p(S)$, and write*

$$-[dN]_{\mathcal{B},\mathcal{B}} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Then

$$a = \mathrm{II}_{\mathbf{q}}(\mathbf{v}_1), \qquad d = \mathrm{II}_{\mathbf{q}}(\mathbf{v}_2), \qquad b = \frac{1}{2}\Big[\mathrm{II}_{\mathbf{q}}(\mathbf{v}_1+\mathbf{v}_2) - \mathrm{II}_{\mathbf{q}}(\mathbf{v}_1) - \mathrm{II}_{\mathbf{q}}(\mathbf{v}_2)\Big].$$

Proof. From the matrix representation, $-dN(\mathbf{v}_1) = a\mathbf{v}_1 + b\mathbf{v}_2$ and $-dN(\mathbf{v}_2) = b\mathbf{v}_1 + d\mathbf{v}_2$. Then by definition of II:

$$\mathrm{II}_{\mathbf{q}}(\mathbf{v}_1) = \langle a\mathbf{v}_1 + b\mathbf{v}_2, \mathbf{v}_1\rangle = a, \qquad \mathrm{II}_{\mathbf{q}}(\mathbf{v}_2) = \langle b\mathbf{v}_1 + d\mathbf{v}_2, \mathbf{v}_2\rangle = d.$$

For b, expand:

$$\begin{aligned}
\mathrm{II}_{\mathbf{q}}(\mathbf{v}_1+\mathbf{v}_2) &= \langle -dN(\mathbf{v}_1+\mathbf{v}_2),\ \mathbf{v}_1+\mathbf{v}_2\rangle \\
&= \langle -dN(\mathbf{v}_1), \mathbf{v}_1\rangle + \langle -dN(\mathbf{v}_1), \mathbf{v}_2\rangle + \langle -dN(\mathbf{v}_2), \mathbf{v}_1\rangle + \langle -dN(\mathbf{v}_2), \mathbf{v}_2\rangle \\
&= \mathrm{II}_{\mathbf{q}}(\mathbf{v}_1) + \mathrm{II}_{\mathbf{q}}(\mathbf{v}_2) + 2b.
\end{aligned}$$

Solving for b gives the stated formula. □

3.5 Normal Curvature

Definition 3.12 (Normal Curvature)**.** Let $\alpha(s)$ be a curve parameterised by arc length on a surface S, passing through $\mathbf{p} = \alpha(0)$. The **normal curvature** is

$$k_n = \kappa\cos\theta,$$

where θ is the angle between $\alpha''(0)$ and $N(\mathbf{p})$.

The normal curvature is the projection of the curvature vector $\alpha''(0)$ onto the surface normal. In particular, $|k_n(\mathbf{p})| \leq \kappa$.

Proposition 3.13 (Normal curvature = second fundamental form)**.** *Let $\alpha(s)$ be a curve parameterised by arc length, and $\mathbf{p} = \alpha(0) \in S$. Then*

$$\mathrm{II}_{\mathbf{p}}(\alpha'(0)) = -\langle dN(\alpha'(0)),\ \alpha'(0)\rangle = k_n(\mathbf{p}).$$

Proof. Since $N(\alpha(s)) \perp \alpha'(s)$, we have $\langle N(\alpha(s)), \alpha'(s)\rangle = 0$. Differentiating:

$$\begin{aligned}
0 &= \frac{d}{ds}\langle N(\alpha(s)), \alpha'(s)\rangle \\
&= \langle dN(\alpha'(s)), \alpha'(s)\rangle + \langle N(\alpha(s)), \alpha''(s)\rangle \\
&= -\mathrm{II}_{\mathbf{p}}(\alpha'(s)) + \langle N(\alpha(s)), \kappa\,\mathbf{n}(s)\rangle \\
&= -\mathrm{II}_{\mathbf{p}}(\alpha'(s)) + \kappa\cos\theta.
\end{aligned}$$

Hence $\mathrm{II}_{\mathbf{p}}(\alpha'(s)) = \kappa\cos\theta = k_n(\mathbf{p})$. □

The significance is that the normal curvature depends only on the unit tangent vector $\alpha'(0)$: if two curves satisfy $\alpha'(0) = \beta'(0)$, they have the same normal curvature.

3.6 Gaussian Curvature

Proposition 3.14. $-dN : T_{\mathbf{p}}(S) \to T_{\mathbf{p}}(S)$ *is a self-adjoint linear transformation.*

Proof. Since dN is self-adjoint, for all $\mathbf{v}, \mathbf{w} \in T_{\mathbf{p}}(S)$:

$$\langle -dN(\mathbf{v}), \mathbf{w}\rangle = -\langle dN(\mathbf{v}), \mathbf{w}\rangle = -\langle \mathbf{v}, dN(\mathbf{w})\rangle = \langle \mathbf{v}, -dN(\mathbf{w})\rangle. \qquad \square$$

Definition 3.15 (Principal curvatures and directions)**.** By the main theorem on self-adjoint maps, there exists an orthonormal basis $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$ of $T_{\mathbf{p}}(S)$ diagonalising $-dN$:

$$-[dN]_{\mathcal{B},\mathcal{B}} = \begin{pmatrix} k_1 & 0 \\ 0 & k_2 \end{pmatrix}.$$

We call $\mathbf{v}_1, \mathbf{v}_2$ the **principal directions** and k_1, k_2 the **principal curvatures** at $\mathbf{p}$.

Proposition 3.16 (Second fundamental form in principal directions)**.** *For* $\mathbf{v} = a\mathbf{v}_1 + b\mathbf{v}_2 \in T_{\mathbf{p}}(S)$,

$$\mathrm{II}_{\mathbf{p}}(\mathbf{v}) = k_1 a^2 + k_2 b^2.$$

Proof.

$$\begin{aligned} \mathrm{II}_{\mathbf{p}}(\mathbf{v}) &= \langle -dN(a\mathbf{v}_1 + b\mathbf{v}_2),\ a\mathbf{v}_1 + b\mathbf{v}_2 \rangle \\ &= \langle k_1 a\mathbf{v}_1 + k_2 b\mathbf{v}_2,\ a\mathbf{v}_1 + b\mathbf{v}_2 \rangle \\ &= k_1 a^2 + k_2 b^2. \end{aligned}$$

□

Proposition 3.17. *The maximal and minimal normal curvatures at* $\mathbf{p}$ *occur at* $\mathbf{v}_1$ *and* $\mathbf{v}_2$ *respectively, and they are perpendicular to each other.*

Definition 3.18 (Gaussian and mean curvatures)**.** Let k_1, k_2 be the principal curvatures at $\mathbf{p}$. The **Gaussian curvature** and **mean curvature** are defined as

$$K(\mathbf{p}) = \det(-[dN_p]_{\mathcal{B},\mathcal{B}}) = k_1 k_2, \qquad H(\mathbf{p}) = \frac{1}{2}\operatorname{tr}(-[dN_p]_{\mathcal{B},\mathcal{B}}) = \frac{k_1 + k_2}{2}.$$

Remarks.

1. $K(\mathbf{p})$ is the determinant of $-dN$ (product of eigenvalues); $2H(\mathbf{p})$ is the trace (sum of eigenvalues).
2. Both $K(\mathbf{p})$ and $H(\mathbf{p})$ are independent of the choice of orthonormal basis.

3.7 Gauss Map in Local Coordinates

In this section we derive formulas for dN, H, and K in terms of the coordinate map $\mathbf{x}$ and the first fundamental form coefficients E, F, G.

Definition 3.19 (Gauss map in local coordinates)**.** We define

$$\mathbf{M}(u, v) := N \circ \mathbf{x}(u, v),$$

viewing the Gauss map in local coordinates. Since $\mathbf{M}_u$ and $\mathbf{M}_v$ lie in $T_p(S)$, which is spanned by $\{\mathbf{x}_u, \mathbf{x}_v\}$, we write

$$\mathbf{M}_u = a_{11}\mathbf{x}_u + a_{21}\mathbf{x}_v, \qquad \mathbf{M}_v = a_{12}\mathbf{x}_u + a_{22}\mathbf{x}_v.$$

Lemma 3.20.

$$d(N \circ \mathbf{x}) = d\mathbf{M} = d\mathbf{x} \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix},$$

which is equivalent to

$$[dN]_{\mathcal{B},\mathcal{B}} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}.$$

Since K and H are determined by $[dN]_{\mathcal{B},\mathcal{B}}$, it remains to express a_{ij} in terms of the first and second fundamental form coefficients.

4 Isometries

4.1 Isometries

Definition 4.1 (Diffeomorphism)**.** A diffeomorphism is a map $\phi : S_1 \to S_2$ such that

- ϕ is smooth (between surfaces),
- ϕ is a bijection (so ϕ^{-1} exists),
- ϕ^{-1} is also smooth (between surfaces).

We say that two surfaces are **diffeomorphic** if there exists a diffeomorphism between them.

Facts.

1. If $\phi : S_1 \to S_2$ is a diffeomorphism, then its inverse $\phi^{-1} : S_2 \to S_1$ is also a diffeomorphism.
2. If $\phi : S_1 \to S_2$ is a diffeomorphism, and $\mathbf{x} : U \to S_1$ is a coordinate function of S_1, then

$$\mathbf{y} = \phi \circ \mathbf{x} : U \to S_2$$

   is a coordinate function of S_2.

Now, we introduce the notion of an **isometry**.

Definition 4.2 (Isometry)**.** A diffeomorphism $\phi : S_1 \to S_2$ is an **isometry** if the arc lengths of tangent vectors remain unchanged under $d\phi : T_{\mathbf{p}}(S_1) \to T_{\phi(\mathbf{p})}(S_2)$ for all $\mathbf{p} \in S_1$, i.e.

$$|\mathbf{u}| = |d\phi(\mathbf{u})| \quad \text{for all } \mathbf{u} \in T_{\mathbf{p}}(S_1),$$

or equivalently, for all $\mathbf{p} \in S_1$ and for all $\mathbf{u} \in T_{\mathbf{p}}(S_1)$,

$$\mathrm{I}_{\mathbf{p}}(\mathbf{u}) = \mathrm{I}_{\phi(\mathbf{p})}(d\phi(\mathbf{u})). \tag{$*$}$$

Definition 4.3 (Isometry, DoCarmo)**.** A diffeomorphism $\phi : S_1 \to S_2$ is an **isometry** if for all $\mathbf{p} \in S_1$ and for all $\mathbf{u}, \mathbf{v} \in T_{\mathbf{p}}(S_1)$,

$$\langle \mathbf{u}, \mathbf{v} \rangle = \langle d\phi(\mathbf{u}), d\phi(\mathbf{v}) \rangle. \tag{$**$}$$

The surfaces S_1, S_2 are then said to be *isometric*.

Remark 4.4.

1. We have an equivalent definition of isometry if we replace Equation ($**$) by the seemingly weaker Equation ($*$). It turns out both are equivalent. We prove this in T9Q2.
2. By definition, an isometry is a diffeomorphism.

4.2 Local Isometries

Definition 4.5 (Local Isometry)**.** A smooth map $\phi : S_1 \to S_2$ is a *local isometry* at $\mathbf{p} \in S_1$ if there is an open neighbourhood V_1 of $\mathbf{p}$ on S_1 and an open neighbourhood V_2 of $\phi(\mathbf{p})$ on S_2 such that $\phi : V_1 \to V_2$ is an isometry.
A smooth map $\phi : S_1 \to S_2$ is a *local isometry* if it is a local isometry at every point on S_1.

Remark 4.6. Key facts about local isometries.

1. Since $\phi : V_1 \to V_2$ is an isometry, it is also a diffeomorphism.
2. Every (global) isometry is a local isometry.
3. A local isometry $\phi : S_1 \to S_2$ is **NOT** necessarily a diffeomorphism. It may not even be a bijection. Hence a local isometry may **NOT** be an isometry.
4. ϕ being a local isometry at a point $\mathbf{p}$ is stronger than saying $\mathrm{I}_{\mathbf{p}}(\mathbf{u}) = \mathrm{I}_{\phi(\mathbf{p})}(d\phi(\mathbf{u}))$ alone. It means $\mathrm{I}_{\mathbf{q}}(\mathbf{u}) = \mathrm{I}_{\phi(\mathbf{q})}(d\phi(\mathbf{u}))$ for all points $\mathbf{q}$ in the open neighbourhood V_1 of $\mathbf{p}$ and for all $\mathbf{u} \in T_{\mathbf{q}}(S_1)$.

Proposition 4.7. *Let $\mathbf{x} : U \to S_1$ and $\mathbf{y} : U \to S_2$ be coordinate functions. Let E_1, F_1, G_1 and E_2, F_2, G_2 denote the coefficients of the first fundamental forms on S_1 and S_2 respectively. Suppose $E_1 = E_2$, $F_1 = F_2$, and $G_1 = G_2$ as functions of $(u, v) \in U$. Then*

$$\Phi = \mathbf{y} \circ \mathbf{x}^{-1} : \mathbf{x}(U) \to \mathbf{y}(U)$$

is an isometry.

Proof. Let $\mathbf{p} = \mathbf{x}(u, v)$. Let $\mathbf{w} = a\mathbf{x}_u + b\mathbf{x}_v = d\mathbf{x}\begin{pmatrix} a \\ b \end{pmatrix}$ be a tangent vector at $\mathbf{p}$. Then,

$$d\Phi(\mathbf{w}) = d\Phi\, d\mathbf{x}\begin{pmatrix} a \\ b \end{pmatrix} = d(\mathbf{y} \circ \mathbf{x}^{-1} \circ \mathbf{x})\begin{pmatrix} a \\ b \end{pmatrix} = a\mathbf{y}_u + b\mathbf{y}_v.$$

Next we compare

$$\mathrm{I}_{\mathbf{p}}(\mathbf{w}) = a^2E_1 + 2abF_1 + b^2G_1,$$
$$\mathrm{I}_{\Phi(\mathbf{p})}(d\Phi(\mathbf{w})) = \mathrm{I}_{\Phi(\mathbf{p})}(a\mathbf{y}_u + b\mathbf{y}_v) = a^2E_2 + 2abF_2 + b^2G_2.$$

Since $E_1 = E_2$, $F_1 = F_2$, $G_1 = G_2$, the two first fundamental forms agree. Since $\Phi : \mathbf{x}(U) \to \mathbf{y}(U)$ is a diffeomorphism, it is an isometry. □

Corollary 4.8. *Let $\phi : S_1 \to S_2$ be a diffeomorphism. Let $\mathbf{x} : U \to S_1$ be a coordinate function. Then $\mathbf{y} = \phi \circ \mathbf{x} : U \to S_2$ is a coordinate function on S_2. Let E_i, F_i, G_i be the coefficients of the first fundamental forms on S_i for $i = 1, 2$. Then the following are equivalent:*

1. *$E_1 = E_2$, $F_1 = F_2$, $G_1 = G_2$ as functions of $(u, v) \in U$.*
2. *$\phi = \mathbf{y} \circ \mathbf{x}^{-1} : \mathbf{x}(U) \to \mathbf{y}(U)$ is an isometry.*

4.3 Gauss Theorema Egregium

Theorem 4.9 (Theorema Egregium)**.** *Let $\phi : S_1 \to S_2$ be a local isometry. Let $p_1 \in S_1$ and $p_2 \in S_2$. Let K and $\bar{K}$ be the Gaussian curvatures at p_1 and p_2 respectively. Then $K = \bar{K}$.*

Proof. The main idea of the proof is as follows:

1. Suppose $\phi : S_1 \to S_2$ is a local isometry.
2. Write the Gaussian curvature K on S_1 in terms of the first fundamental form of S_1.
3. Similarly $\overline{K}$ on S_2 can be written in terms of the first fundamental form of S_2.
4. Since ϕ is a local isometry, the first fundamental forms agree locally.
5. Hence $K = \overline{K}$.

Let S be a regular surface and $\mathbf{x} : U \to S$ a coordinate function. The partial derivatives $\mathbf{x}_{uu}, \mathbf{x}_{uv} = \mathbf{x}_{vu}, \mathbf{x}_{vv}$ can be expressed in the basis $\{\mathbf{x}_u, \mathbf{x}_v, N\}$:

$$\begin{aligned} \mathbf{x}_{uu} &= \Gamma^1_{11}\mathbf{x}_u + \Gamma^2_{11}\mathbf{x}_v + L_1 N, & \mathbf{x}_{uv} &= \Gamma^1_{12}\mathbf{x}_u + \Gamma^2_{12}\mathbf{x}_v + L_2 N, \\ \mathbf{x}_{vu} &= \Gamma^1_{21}\mathbf{x}_u + \Gamma^2_{21}\mathbf{x}_v + \bar{L}_2 N, & \mathbf{x}_{vv} &= \Gamma^1_{22}\mathbf{x}_u + \Gamma^2_{22}\mathbf{x}_v + L_3 N, \\ N_u &= a_{11}\mathbf{x}_u + a_{21}\mathbf{x}_v, & N_v &= a_{12}\mathbf{x}_u + a_{22}\mathbf{x}_v. \end{aligned}$$

The coefficients Γ^k_{ij} are the **Christoffel symbols**.
Lemma 4. The Christoffel symbols Γ^k_{ij} are functions of E, F, G and their partial derivatives. In particular,

$$\Gamma^1_{11} = \frac{GE_u - 2FF_u + FE_v}{2(EG - F^2)}, \qquad \Gamma^2_{11} = \frac{-FE_u + 2EF_u - EE_v}{2(EG - F^2)}.$$

Lemma 6.

$$\begin{aligned} EK &= (\Gamma^2_{11})_v - (\Gamma^2_{12})_u + \Gamma^1_{11}\Gamma^2_{12} + \Gamma^2_{11}\Gamma^2_{22} - \Gamma^1_{12}\Gamma^2_{11} - (\Gamma^2_{12})^2, \\ FK &= (\Gamma^1_{12})_u - (\Gamma^1_{11})_v + \Gamma^1_{12}\Gamma^2_{12} - \Gamma^2_{11}\Gamma^1_{22}. \end{aligned}$$

The proof starts from $(\mathbf{x}_{uu})_v - (\mathbf{x}_{uv})_u = 0$, substitutes the expressions above, applies the product rule and the Weingarten equations, then collects coefficients in $\{\mathbf{x}_u, \mathbf{x}_v, N\}$.
Completing the proof. Let $\mathbf{x} : U \to S_1$ be a coordinate function; then $\mathbf{y} = \phi \circ \mathbf{x} : U \to S_2$ is also a coordinate function. Since ϕ is a local isometry, $E = \overline{E}$, $F = \overline{F}$, $G = \overline{G}$ (and all their partial derivatives agree), so $\Gamma^k_{ij} = \overline{\Gamma}^k_{ij}$. By Lemma 6, K depends only on E, F, G and the Christoffel symbols, hence $K = \overline{K}$. □

Remark 4.10. A modification of the proof of Lemma 6 yields the **Mainardi–Codazzi equations**:

$$\begin{aligned} e_v - f_u &= e\Gamma^1_{12} + f(\Gamma^2_{12} - \Gamma^1_{11}) - g\Gamma^2_{11}, \\ f_v - g_u &= e\Gamma^1_{22} + f(\Gamma^2_{22} - \Gamma^1_{12}) - g\Gamma^2_{12}. \end{aligned}$$

Theorem 4.11 (Fundamental Theorem of Surfaces)**.** *Let E, F, G, e, f, g be smooth functions on an open set U with $EG - F^2 > 0$, $E > 0$, $G > 0$. Form the Christoffel symbols Γ^k_{ij} from E, F, G as in Lemma 4. Suppose E, F, G, e, f, g and Γ^k_{ij} satisfy Lemma 6 and the two Mainardi–Codazzi equations. Then for any $(u_0, v_0) \in U$ there exists a regular surface S and a coordinate function $\mathbf{x} : U' \to S$ (with $U' \subseteq U$ open, $(u_0, v_0) \in U'$) such that E, F, G and e, f, g are the first and second fundamental forms of S respectively.*

5 Vector Fields & Geodesics

5.1 Tangent Vector Fields

Definition 5.1 (Vector Field). Let S be a regular surface and let $W \subseteq S$ be an open subset. A **vector field w** *on* W is a correspondence that assigns to each $p \in W$ a tangent vector $\mathbf{w}(p) \in T_p(S)$.
We can express $\mathbf{w}$ in terms of a coordinate function. More precisely, if $p = \mathbf{x}(u, v) \in S$, then

$$\mathbf{w}(\mathbf{x}(u, v)) = a(u, v)\,\mathbf{x}_u + b(u, v)\,\mathbf{x}_v \;\in\; T_p(S)$$

for some functions a, b on U. In matrix form,

$$\mathbf{w}(p) = a(u, v)\,\mathbf{x}_u + b(u, v)\,\mathbf{x}_v = d\mathbf{x}\begin{pmatrix} a(u, v) \\ b(u, v) \end{pmatrix}.$$

We interpret this expression by viewing $\begin{pmatrix} a(u, v) \\ b(u, v) \end{pmatrix}$ as a vector field on U, and $d\mathbf{x}$ as the map transforming it to the field $\mathbf{w}$ on S.

Definition 5.2 (Smooth Vector Field). We say that a vector field $\mathbf{w}$ is **smooth** at $p_0 = \mathbf{x}(u_0, v_0)$ if the vector field $\begin{pmatrix} a(u, v) \\ b(u, v) \end{pmatrix}$ on U is smooth at (u_0, v_0).

This definition is independent of the choice of coordinate function.

Proposition 5.3 (Smoothness is independent of coordinate function). *The smoothness of* $\mathbf{w}$ *at a point* p *does not depend on the choice of coordinate function used to define it.*

Proof. Suppose $\mathbf{y}\colon U' \to S$ is another coordinate function with $p \in \mathbf{y}(U')$. Then

$$\mathbf{w}(q) = a(u, v)\,\mathbf{x}_u + b(u, v)\,\mathbf{x}_v = c(\xi, \eta)\,\mathbf{y}_\xi + d(\xi, \eta)\,\mathbf{y}_\eta$$

for all q in a neighbourhood of p. Without loss of generality assume $\mathbf{x}(U) = \mathbf{y}(U')$ (otherwise restrict to $U \cap U'$). Suppose $\mathbf{w}$ is smooth with respect to $\mathbf{y}$. One then shows via the chain rule and the transition map $\mathbf{y}^{-1} \circ \mathbf{x}$ that it is also smooth with respect to $\mathbf{x}$. $\square$

5.2 A Good Definition for the Derivative of a Vector Field

Definition 5.4 (Differentiation along a curve)**.** Let $\alpha(t) = \mathbf{x}(u(t), v(t))$ be a curve on S. Write

$$\mathbf{w}(\alpha(t)) = a(u(t), v(t))\,\mathbf{x}_u + b(u(t), v(t))\,\mathbf{x}_v.$$

If $\alpha(0) = p$ and $\alpha'(0) = (a_1, a_2)$, then by the chain rule,

$$\frac{d}{dt}\mathbf{w}(\alpha(t))\bigg|_{t=0} = a_1 \frac{\partial \mathbf{w}}{\partial u}\bigg|_p + a_2 \frac{\partial \mathbf{w}}{\partial v}\bigg|_p.$$

Crucially, this expression depends on the curve α *only through its tangent vector* $\alpha'(0) = (a_1, a_2)$.

Differentiating along the line $\alpha(t) = p + t\mathbf{v} = (p_1 + ta_1,\ p_2 + ta_2)$ recovers the classical directional derivative:

$$\frac{d}{dt}\mathbf{w}(\alpha(t))\bigg|_{t=0} = \lim_{\Delta t \to 0} \frac{\mathbf{w}(p_1 + a_1\Delta t,\ p_2 + a_2\Delta t) - \mathbf{w}(p_1, p_2)}{\Delta t}.$$

In general, $\frac{d}{dt}\mathbf{w}(\alpha(t))$ need *not* lie in the tangent plane. This motivates the following definition.

Definition 5.5 (Covariant Derivative)**.** The **covariant derivative** of $\mathbf{w}$ at p relative to the tangent vector $\alpha'(0)$ is

$$\frac{D\mathbf{w}}{dt} = \frac{d}{dt}\mathbf{w}(\alpha(t)) - \left\langle \frac{d}{dt}\mathbf{w}(\alpha(t)),\, \mathbf{N} \right\rangle \mathbf{N},$$

i.e. the orthogonal projection of $\frac{d}{dt}\mathbf{w}(\alpha(t))$ onto the tangent plane $T_p(S)$.

5.3 Christoffel Symbols in Vector Fields

Christoffel Symbols in the Covariant Derivative

Let $\alpha(t) = (u(t), v(t))$ and write

$$\mathbf{w}(\alpha(t)) = A(t)\,\mathbf{x}_u + B(t)\,\mathbf{x}_v, \qquad A(t) = a(u(t), v(t)), \quad B(t) = b(u(t), v(t)).$$

Differentiating by the product rule and expanding via the Christoffel symbols, then projecting onto the tangent plane, yields

$$\begin{aligned}\frac{D\mathbf{w}}{dt} = &\Big(A' + \Gamma^1_{11}Au' + \Gamma^1_{12}Av' + \Gamma^1_{21}Bu' + \Gamma^1_{22}Bv'\Big)\mathbf{x}_u \\ &+ \Big(B' + \Gamma^2_{11}Au' + \Gamma^2_{12}Av' + \Gamma^2_{21}Bu' + \Gamma^2_{22}Bv'\Big)\mathbf{x}_v.\end{aligned}$$

Remark 5.6. Since the Christoffel symbols Γ^k_{ij} are determined entirely by the first fundamental form coefficients E, F, G (as shown in Chapter 4), the covariant derivative is an *intrinsic* quantity: it can be computed purely from the first fundamental form, without any reference to the unit normal $\mathbf{N}$.

Maps Between Vector Fields

Let S_1, S_2 be regular surfaces, $\mathbf{x}\colon U \to S_1$ a coordinate function, and $\phi\colon S_1 \to S_2$ a diffeomorphism. Then $\mathbf{y} = \phi \circ \mathbf{x}\colon U \to S_2$ is also a coordinate function. For a vector field $\mathbf{w} = a(u, v)\,\mathbf{x}_u + b(u, v)\,\mathbf{x}_v$ on S_1,

$$d\phi(\mathbf{w}) = a(u, v)\,\mathbf{y}_u + b(u, v)\,\mathbf{y}_v,$$

which is a vector field on S_2.

Proposition 5.7 (Isometries and vector fields)**.** *Let $\phi\colon S_1 \to S_2$ be an isometry. Then*

$$\frac{D\big(d\phi(\mathbf{w})\big)}{dt} = d\phi\left(\frac{D\mathbf{w}}{dt}\right).$$

Proof (sketch). Let E_i, F_i, G_i denote the first fundamental form coefficients with respect to $\mathbf{x}$ on S_1 and $\mathbf{y} = \phi \circ \mathbf{x}$ on S_2. Since ϕ is an isometry, the two first fundamental forms coincide, hence the Christoffel symbols of S_1 and S_2 are identical. The covariant derivative formula above then shows both sides have the same expression in coordinates. □

5.4 Geodesic Curvature

In this section, we introduce the *geodesic curvature*.

Consider a point $\mathbf{p} = \alpha(s_0) \in S$ for some point $s_0 \in U$. Then, we define

$$\mathbf{t} = \alpha'(s) \qquad \mathbf{s} = N \wedge \mathbf{t}$$

and these two (unit) vectors form an *orthonormal basis* at $\mathbf{p}$.

And then, we have $\alpha''(s_0)$ to be perpendicular to $\mathbf{t}$, but not necessarily on the tangent plane. So, it is made of components of N and $\mathbf{s}$.

It is easier to visualise with this diagram.

insert diagram here.

Then, we have:

$$\alpha''(s_0) = \langle N, \alpha''(s_0)\rangle N + \langle \mathbf{s}, \alpha''(s)\rangle \mathbf{s}$$

And $\langle N, \alpha''(s_0)\rangle$ is the size of the vector that is projected from $\alpha''(s_0)$ onto the "normal plane", which we recall from 3.12 is in fact the *normal curvature*!

Proposition 5.8. *Let k_n be the normal curvature at $\mathbf{v}$. We show that the following are equivalent:*

1. $k_n = \langle N, \alpha''(0)\rangle$
2. $k_n = \mathrm{II}_v(v)$

Proof. We know that $\langle N(\alpha(s)), \alpha'(s)\rangle = 0$ since they are perpendicular to each other. So, it follows that

$$\begin{aligned}\langle N(\alpha(s)), \alpha'(s)\rangle = 0 &\iff \frac{d}{ds}\langle N(\alpha(s)), \alpha'(s)\rangle = 0 \\ &\iff \langle N, \alpha''(s)\rangle + \langle dN(\alpha'(s)), \alpha'(s)\rangle = 0 \\ &\iff \langle N, \alpha''(s)\rangle = -\langle dN(\alpha'(s)), \alpha'(s)\rangle = \mathrm{II}_{\mathbf{p}}(\alpha'(0)) = k_n\end{aligned}$$

□

And we define the second component as the geodesic curvature.

Definition 5.9 (Geodesic Curvature)**.** We define

$$k_g := \langle \mathbf{s}, \alpha''(s_0)\rangle$$

to be the *geodesic curvature* of $\alpha(s)$ at the point s_0.

And so, we can simplify the expression for α'' to be

$$\alpha''(s_0) = k_n\mathbf{N} + k_g\mathbf{s}$$

Proposition 5.10. *Let k, k_n and k_g be the curvature, normal curvature and geodesic curvature of α respectively. Then,*

$$k = k_n^2 + k_g^2$$

Proof. Follows from Pythagroas Theorem. □

5.5 Geodesic

We introduce the concept of a geodesic. First, we motivate the idea using a proposition.

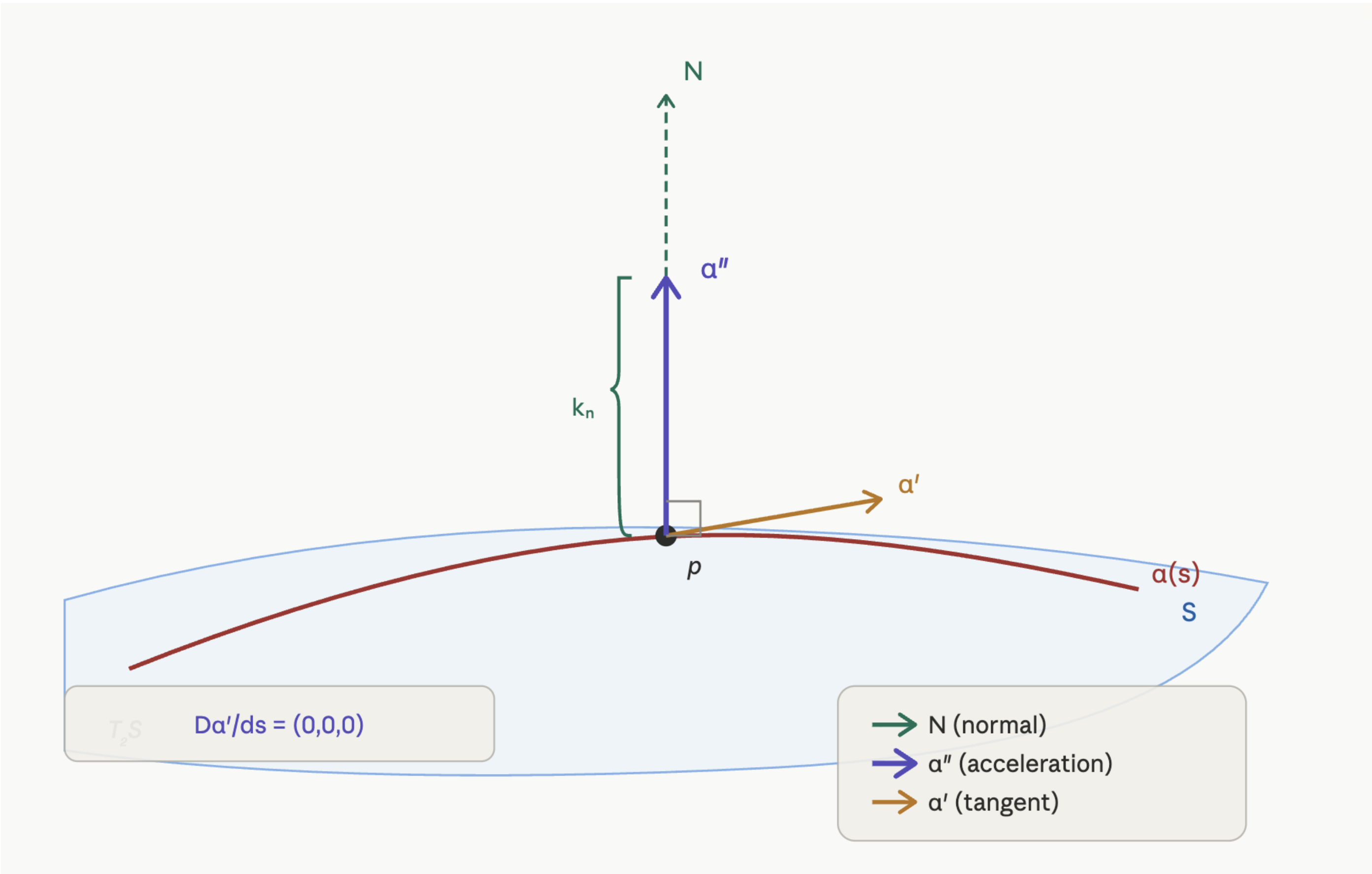
Proposition 5.11. *Let $\alpha(s)$ be a curve on a surface S parameterized by arc length. Then, the following are equivalent. For all s,*

1. $$\frac{D\alpha'}{ds} = (0, 0, 0)$$
2. *$\alpha''(s)$ is perpendicular to the tangent space.*
3. *$\alpha''(s)$ is parallel to the normal vector $\mathbf{N}$.*
4. *The geodesic curvature k_g of α is zero.*

Proof. (1) $\Leftrightarrow$ (2) : $\frac{D\alpha'}{ds}$ is the orthogonal projection of α''. So, it is clear that it is 0 if and only if α'' is perpendicular to the tangent space.

(2) $\Leftrightarrow$ (3) : The tangent space is perpendicular to $\mathbf{N}$. So, α'' must be parallel to $\mathbf{N}$.

(3) $\Leftrightarrow$ (4) : k_g is the *length* of the projection of α'' onto the tangent plane, which is 0 because α'' is parallel to $\mathbf{N}$ by assumption. And the converse direction follows by a reverse argument. □



This is a visualisation of the following proposition. So, α'' is perpendicular to the tangent space, so $k_g = 0$.

Definition 5.12 (Geodesic)**.** A curve *parameterised by arc length* satisfying any of the four conditions in 5.11 is called a **geodesic**.

Now, why is the definition of a geodesic as such? Well, how I interpret it is that given a curve between a and b, the shortest path would be the one where the α'' does not go left or right, so it just goes straight, following the line of the normal N — "the curve outlines the boundary of the surface".

Lastly, we have a proposition linking geodesics to isometry.

Proposition 5.13 (Geodesics are preserved by isometry)**.** *Let $\phi : S_1 \to S_2$ be an isometry. And suppose $\alpha(s)$ is a geodesic on S_1. Then $\beta(s) = \phi(\alpha(s))$ is a geodesic on S_2.*

Proof. First, we check that $\beta(s)$ is parameterized by arc length. Since ϕ is an isometry,

$$|\beta'(s)| = |d\phi(\alpha'(s))| = |\alpha'(s)| = 1$$

The first equality is using chain rule, and the second equality is by definition of an isometry (4.2). Hence, β is indeed parameterized by arc length.
Next, let $\mathbf{w} = \alpha'(s)$. The curve $\alpha(s)$ being a geodesic means (by 5.9) that $\frac{D\mathbf{w}}{ds} = (0, 0, 0)$. And so using Proposition 5.7,

$$\frac{D\beta'}{ds} = \frac{D(d\phi(\mathbf{w}))}{ds} = d\phi\left(\frac{D\mathbf{w}}{ds}\right) = d\phi(0, 0, 0) = (0, 0, 0)$$

So β is indeed also a geodesic. □

5.6 Existence of Geodesics

Note: I combined 5.7 and 5.8 into this chapter.

In this section, we ask the question: *Given any surface, can we always find a geodesic?*

Proposition 5.14. *Let $\mathbf{p} \in S$ be a point on a surface and let $\mathbf{v}$ be a unit tangent vector at $\mathbf{p}$. Then there exists a unique geodesic $\alpha(s)$ parameterized by arc length which passes through $\mathbf{p}$ at $s = 0$ and $\alpha'(0) = \mathbf{v}$.*

Proof. Skip. See notes for the full proof. □

Proposition 5.15. *Given two points **p**,**q** on a regular surface S which are sufficiently near to each other. Then*

*(i) There is a geodesic $\alpha(s)$ on the surface joining **p** and **q**.*

*(ii) If $\beta(s)$ is another curve on the surface joining **p**,**q**, then the arc length of $\beta(s)$ must be same or less than that of $\alpha(s)$.*

5.7 The Exponential Map

Proposition 5.16. *Let $\alpha(s)$ be a geodesic. Then, given a constant $c > 0$, it follows that $\gamma(t) = \alpha(ct)$ is also a geodesic.*

Proof. We verify that the geodesic curvature of γ vanishes. By the chain rule,

$$\gamma'(t) = c\alpha'(ct), \qquad \gamma''(t) = c^2\alpha''(ct).$$

Since α is a geodesic, $\alpha''(s)$ is parallel to the normal $\mathbf{N}$ for all s. In particular, $\alpha''(ct)$ is parallel to $\mathbf{N}$ at $\alpha(ct) = \gamma(t)$, and so $\gamma''(t) = c^2\alpha''(ct)$ is also parallel to $\mathbf{N}$. Hence the geodesic curvature of γ is

$$k_g(\gamma) = 0,$$

and so γ is a geodesic. □

Now, we have a few important stuff to get through. First,

Theorem 5.17. *Give any point $\mathbf{p}$ on a regular surface S, and a nonzero vector $\mathbf{v} \in T_{\mathbf{p}}(S)$, there exists a unique geodesic $\gamma : (-\epsilon, \epsilon) \to S$ with $\gamma(0) = \mathbf{p}$ and $\gamma'(0) = \mathbf{v}$. We denote this geodesic by $\gamma(t, \mathbf{v})$.*

Proof. Follows from (5.14). □

This theorem essentially says that given a point $\mathbf{p}$ on a regular surface, we can find a geodesic with tangent vector $\mathbf{v}$. It is essentially the same as 5.14, but now we don't need the condition that $\mathbf{v}$ is a unit vector.

Theorem 5.18. *Let S be a regular surface and let $\mathbf{p} \in S$. Then, there exists $\epsilon > 0$ such that the map*

$$\begin{aligned}\gamma : (-2, 2) \times B(\epsilon) &\to S \\ (t, \mathbf{v}_\epsilon) &\mapsto \gamma(t, \mathbf{v}_\epsilon)\end{aligned}$$

is a smooth map. Here, $B(\epsilon) := \{\mathbf{v} \in T_{\mathbf{p}}(S) : |\mathbf{v}| < \epsilon\}$.

This theorem is very unintuitive, but let's try to understand it. Basically, $B(\epsilon)$ is the set of all possible tangent vectors of length less than some small ϵ. Then, this theorem says that if we *perturb* $\mathbf{v}$ slightly, the geodesic will not change drastically, and in fact, will change *smoothly*.

Example 5.19. We provide an example to demonstrate the notion of smoothness of γ. Consider the plane $S = \mathbb{R}^2$ and the point $p = (0, 0)$. A geodesic in $\mathbb{R}^2$ is just a straight line, so given a tangent vector $\mathbf{v} = (a, b)$, the geodesic is simply

$$\gamma(t, \mathbf{v}) = t\mathbf{v} = (ta, tb).$$

The exponential map is then

$$\exp_p(\mathbf{v}) = \gamma(1, \mathbf{v}) = (a, b).$$

Now suppose we perturb $\mathbf{v}$ slightly, replacing it with $\mathbf{v}_\epsilon = (a + \epsilon, b)$ for some small $\epsilon > 0$. Then

$$\gamma(t, \mathbf{v}_\epsilon) = (t(a + \epsilon),\ tb).$$

The difference between the two geodesics at any time t is

$$\|\gamma(t, \mathbf{v}_\epsilon) - \gamma(t, \mathbf{v})\| = |t\epsilon| \leq |\epsilon|.$$

So as $\epsilon \to 0$, the perturbed geodesic converges uniformly to the original one. This is what smoothness of $\gamma(t, \mathbf{v})$ in $\mathbf{v}$ means: a small change in the initial velocity produces a small change in the entire curve, with no sudden jumps.

We are now ready to define the exponential map.

Definition 5.20 (Exponential Map)**.** Let $\mathbf{v} \in B(\epsilon)$. We define

$$\exp_{\mathbf{p}}(\mathbf{v}) := \gamma(1, \mathbf{v}) \in S,$$

that is, the point obtained by following the geodesic starting at $\mathbf{p}$ with initial velocity $\mathbf{v}$, for time $t = 1$.

Think of $\exp_{\mathbf{p}}(\mathbf{v})$ as: *stand at* $\mathbf{p}$, *face direction* $\mathbf{v}$, *walk in a straight line (on the surface) for distance* $|\mathbf{v}|$, *and see where you end up.* The arc length from $\mathbf{p}$ to $\exp_{\mathbf{p}}(\mathbf{v})$ along the geodesic is exactly $|\mathbf{v}|$, so the length of the tangent vector encodes the distance travelled.

Now, we want exp to be a coordinate map, so we want it to be a *diffeomorphism*.

Proposition 5.21. *Let* $\mathbf{p} \in S$. *There exists* $\epsilon > 0$ *such that*

$$\exp_{\mathbf{p}} : B(\epsilon) \to S$$

is defined and smooth.

Proof. Immediate corollary of Theorem **??**. □

Proposition 5.22. *Given* $\exp_{\mathbf{p}}$ *as in Proposition 5.21, there exists an open neighbourhood* $U \subset B(\epsilon)$ *of the origin* $0 \in T_{\mathbf{p}}(S)$ *such that*

$$\exp_{\mathbf{p}} : U \to S$$

is a diffeomorphism onto its image $V := \exp_{\mathbf{p}}(U) \subseteq S$.

Proof. By Proposition 5.21, $\exp_{\mathbf{p}}$ is smooth. It remains to show that its differential

$$d(\exp_{\mathbf{p}})_0 : T_0(B(\epsilon)) \to T_{\mathbf{p}}(S)$$

is nonsingular at 0, so that the inverse function theorem applies.
We identify $T_0(B(\epsilon))$ with $T_{\mathbf{p}}(S)$ in the natural way. Let $\mathbf{v} \in T_{\mathbf{p}}(S)$ and consider the straight line $\alpha(t) = t\mathbf{v}$ in $B(\epsilon)$. Define

$$\beta(t) = (\exp_{\mathbf{p}} \circ \alpha)(t) = \exp_{\mathbf{p}}(t\mathbf{v}) = \gamma(t, \mathbf{v}),$$

which is the geodesic on S starting at $\mathbf{p}$ with velocity $\mathbf{v}$. Differentiating at $t = 0$,

$$\beta'(0) = \frac{d}{dt}\Big|_{t=0} \gamma(t, \mathbf{v}) = \mathbf{v}.$$

Hence $d(\exp_{\mathbf{p}})_0(\mathbf{v}) = \mathbf{v}$ for all $\mathbf{v} \in T_{\mathbf{p}}(S)$, so $d(\exp_{\mathbf{p}})_0$ is the identity map, which is nonsingular. By the inverse function theorem, $\exp_{\mathbf{p}}$ is invertible on some open neighbourhood U of 0, with smooth inverse. Hence it is a diffeomorphism. □

The key insight in the proof is that $d(\exp_{\mathbf{p}})_0$ is just the *identity* — pushing a tangent vector $\mathbf{v}$ through the exponential map (to first order) just gives $\mathbf{v}$ back. This makes intuitive sense: very close to $\mathbf{p}$, the surface looks flat, so the exponential map looks like the identity.

Proposition 5.22 says that $\exp_{\mathbf{p}}$ gives a valid coordinate system near $\mathbf{p}$. That is, every point q sufficiently close to $\mathbf{p}$ can be uniquely described by a tangent vector $\mathbf{v}$ at $\mathbf{p}$, via $q = \exp_{\mathbf{p}}(\mathbf{v})$.

The open set $V = \exp_{\mathbf{p}}(U)$ is called a **normal neighbourhood** of $\mathbf{p}$.
x

5.8 Geodesic Coordinate Functions

Recall from Proposition 5.22 that there exists an open neighbourhood $U \subset T_{\mathbf{p}}(S)$ of the origin such that

$$\exp_{\mathbf{p}} : U \to V := \exp_{\mathbf{p}}(U)$$

is a diffeomorphism onto its image. The set V is an open subset of S containing $\mathbf{p}$, and is called a **normal neighbourhood** of $\mathbf{p}$.

This allows us to construct coordinate systems on the surface using the tangent plane at $\mathbf{p}$.

Definition 5.23 (Normal coordinates)**.** Let $\mathbf{p} \in S$. Choose two orthonormal vectors $\mathbf{e}_1, \mathbf{e}_2 \in T_{\mathbf{p}}(S)$.
Every vector $\mathbf{w} \in U$ can be written uniquely as

$$\mathbf{w} = u\mathbf{e}_1 + v\mathbf{e}_2.$$

We define a coordinate map

$$\mathbf{x}(u, v) = \exp_{\mathbf{p}}(u\mathbf{e}_1 + v\mathbf{e}_2).$$

If $S \ni \mathbf{q} = \mathbf{x}(u, v)$, we say that (u, v) are the **normal coordinates** of q.

Proposition 5.24. *Let (u, v) be normal coordinates centred at $\mathbf{p}$. Given*

$$E = \langle x_u, x_u \rangle, \qquad F = \langle x_u, x_v \rangle, \qquad G = \langle x_v, x_v \rangle$$

to be the coefficients of the first fundamental form, then

$$E(\mathbf{p}) = G(\mathbf{p}) = 1, \qquad F(\mathbf{p}) = 0.$$

Thus, at the point $\mathbf{p}$ the coordinate system behaves like the standard Euclidean coordinate system: the coordinate curves are orthogonal and have unit length. This reflects the fact that a surface looks flat when viewed infinitesimally.

Definition 5.25 (Geodesic polar coordinates)**.** Let (ρ, θ) be polar coordinates in the tangent plane $T_{\mathbf{p}}(S)$. We define

$$w(\rho, \theta) = \rho(\cos\theta\, \mathbf{e}_1 + \sin\theta\, \mathbf{e}_2)$$

and

$$x(\rho, \theta) = \exp_{\mathbf{p}}(w(\rho, \theta)).$$

The coordinates (ρ, θ) are called **geodesic polar coordinates**.

Geodesic polar coordinates are analogous to polar coordinates in the plane. The parameter ρ represents the geodesic distance from $\mathbf{p}$, while θ specifies the direction of the geodesic leaving $\mathbf{p}$.

Proposition 5.26. *Let (ρ, θ) be geodesic polar coordinates and let*

$$E = \langle x_\rho, x_\rho \rangle, \qquad F = \langle x_\rho, x_\theta \rangle, \qquad G = \langle x_\theta, x_\theta \rangle.$$

Then

$$E = 1, \qquad F = 0,$$

and

$$\lim_{\rho \to 0} \sqrt{G} = 0, \qquad \lim_{\rho \to 0} (\sqrt{G})_\rho = 1.$$

The condition $E = 1$ means that radial curves have unit speed, while $F = 0$ means that radial geodesics are orthogonal to curves with constant ρ (the geodesic circles centred at $\mathbf{p}$).

Definition 5.27. A coordinate map $x(u, v)$ is called an **orthogonal coordinate system** if

$$F(u, v) = \langle x_u, x_v \rangle = 0$$

for all (u, v).

Theorem 5.28 (Gauss Lemma)**.** *Let $\mathbf{p}$ be a point on a regular surface S. Then there exists a coordinate system around $\mathbf{p}$ such that*

$$\langle x_\rho, x_\theta \rangle = 0.$$

In other words, the radial geodesics issuing from $\mathbf{p}$ are perpendicular to the geodesic circles centred at $\mathbf{p}$.

Gauss Lemma generalises the familiar fact from Euclidean geometry that radial lines are perpendicular to circles. It plays an important role in many geometric results, including the Gauss–Bonnet theorem and the study of minimal properties of geodesics.

5.9 Geodesics and Calculus of Variations

— not tested —

6 Gauss-Bonnet Theorem

– not tested –

7 Tutorials

8 Tutorials

9 Cheatsheet